(Yet another) definition of reproducibility and replicability

SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

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Housekeeping rules

- Please **adhere to** SORTEE Code of Conduct
- **Ask anything**: “raise hand” 👋 or use the chat anytime!
- Feel free to **comment & answer each other’s questions** in the chat.
- **Write in the chat**: name, location, favorite ice-cream flavor 🍦

Many definitions...

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

Image credit: “The Turing Way” online handbook

…with overlapping meanings…

From the Cambridge English Dictionary:

- Reproduce: “to produce a copy of something”; “to show or do something again”.

- Replicate: “to make or do something again in exactly the same way”; “to copy or repeat something”.

...stressing different aspects

- iRISE definition
- Devezer's
- ...?

Discussion

- What other definitions do you know?
- Which one(s) do you use in your practice?
- What problems do you find when using them?

So… what's the matter?

- No just "computational"
- Ambiguous use (e.g. *Reproducibility Project*)
- Confusion

A key distinction

iRISE definitions:

- Replicability:

> “the extent to which design, implementation, analysis, and reporting enable a third party to repeat a study”

- Reproducibility:

> “the extent to which results agree with replication studies”

What is different?

Porcess vs. result

Can we carry out (exactly) *the same research process* again?

Can we obtain *results that agree* with the original ones?

Psycholinguistic intermission

Priming effect:

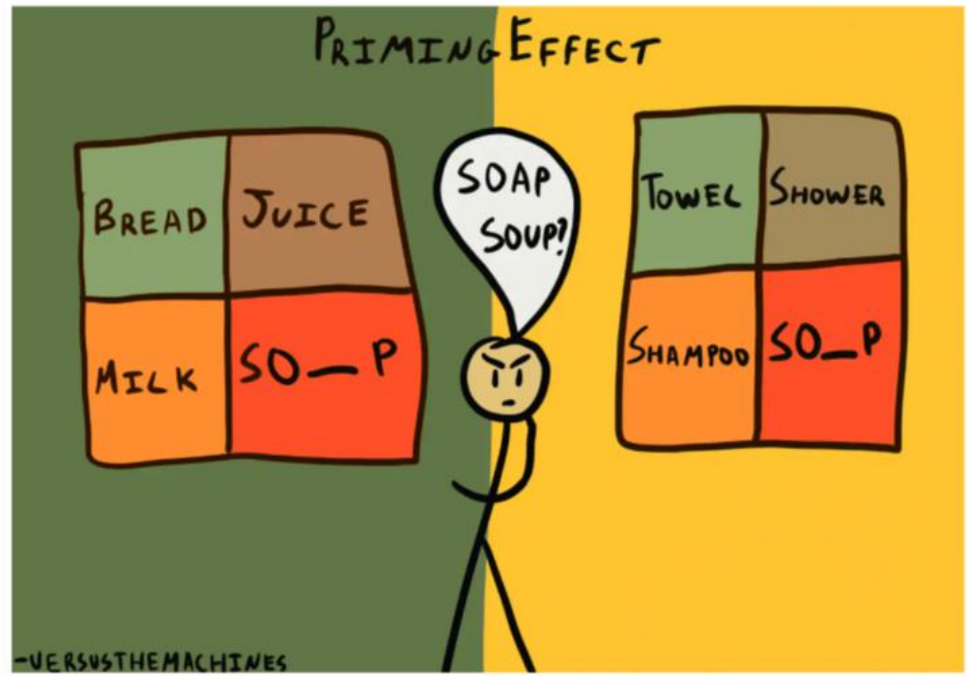


Image credit: The Decision Lab

Psycholinguistic intermission

Syntactic category priming:

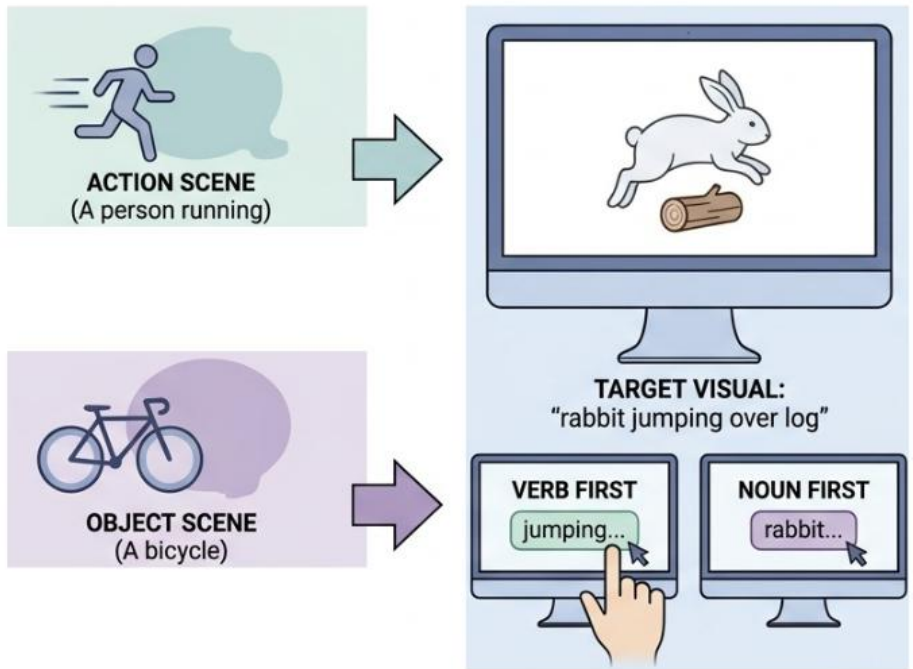


Image credit: Original (with help from Gemini)

Psycholinguistic intermission

A linguistic “intuition”:

- **Reproduce** evokes *action*; “produce” again.
- **Replicate** → transitivity: make a “replica” of something.

Priming of syntactic categories

Proposed definitions:

Reproducibility:

> "the extent to which **research procedure(s)** can be carried out again with *as much fidelity as possible* to the original one(s)"

Replicability:

> "the extent to which original outputs are *compatible* with the new ones that are obtained under *equivalent conditions*"

“as much fidelity as possible”

Is strict reproduction even possible?

- Environmental conditions
- Human factors
- Undocumented conventions
- Context-specific practices

“compatible” outputs

- Conceptually similar effect
- Same decision in NHST
- Same-sign effect (under directional NH rejection)
- Similar prediction (within acceptable deviation)
- A (statistically) non-different size effect
- Exact numerical agreement within machine precision.

Same equivalence class

The connection between them

Reproducibility

↓

conditions for comparison

↓

Replicability

Is reproducibility a **necessary condition** for replicability?

Open questions

- How much fidelity is “enough” fidelity?
- How to quantify “compatible” results?
- How to define equivalence classes?
- Do these definitions reduce confusion?

Take away

Repeat research process ≠ Repeat outputs

Proposal:

- Repeat research process: **Reproduce**
- Repeat outputs: **Replicate**

The end!



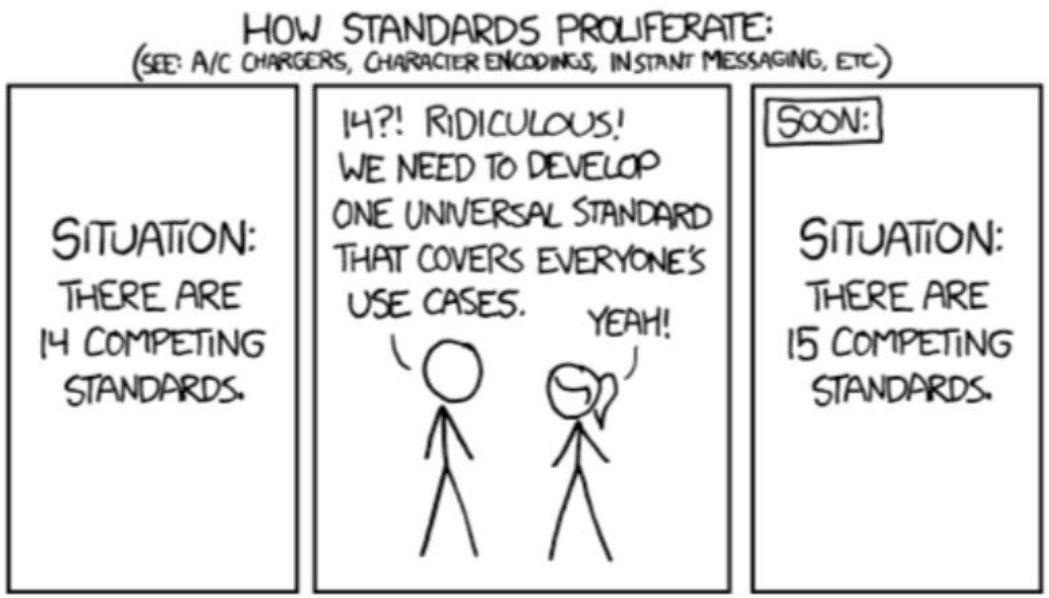


Image credits: Wallpaper Safari / XKCD